Ultrafast X-Ray Experiments

Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21

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1. Einleitung

2. Theoretical Background

2.1. Synchrotronradiation

When accelerating electrons or other charged particles to relativistic speeds along a curved trajectory they emit so-called synchrotron radiation. It gains is name from the ring accelerators, the sychrotrons, where it was first observed. A typical synchrotron facility consists of a linear accelarator leading into a booster ring in which the electrons are brought up to relativistic speeds [1][2]. The electrons are then fed into a storage ring in which they continue to circulate at relativistic speeds. From the storage ring the beam-lines lead tangentially away in which the synchrotron radiation is utilized in experimental hutches [1][2]. Synchrotron radiation is bright X-radiation with a wide tunability in energy, which is useful for examining structures and dynamics of materials, as x-ray photons have energies in the order of several kiloelectronvolts (keV) [2]. This means their wavelengths are in the order of magnitude of Angstroms and comparable to the spacing of atoms in solid materials. This is useful, as it means that atoms in a crystalline array can diffract x-ray beams and the interference patterns can be measured [1].

There are four generations of synchrotrons with the main difference being the brilliance achieved [1]. The first was the bending magnet in the synchrotron as it was used for high energy collider physics with no additional modifications. The second generation introduced the wiggler, a configuration of dipol-magnets with alternating polarity to achieve synchrotron radiation a linear path. The third generation synchrotron radiation facilities were designed for low emittance of the beams with more effective linear magnet configurations, the undulators [2]. A low emittance means a smaller angular divergence of the beam and a higher brilliance [2].

2.1.1. Free Electron Laser (FEL)

Free-electron lasers are coherent tradiation sources that are capable of even higher brilliance than third generation synchrotron facilities [3]. They are in theory able to operate at any arbitrary wavelength or spectrum of wavelength with the limit being the energy and quality of the accelerated electron beam [3]. The duration of a single X-ray pulse can also be reduced to even less than a few femtoseconds. This allows for an investigation of molecular processes and photochemical reactions through a form of sequential snapshots with a characteristic time of under a picosecond [1].

They use a process called self-amplified stimulated emission (SASE). SASE free-electron lasers typically use monochromatic electron beams with a long, high-quality undulator around the order of magnitude of a few hundert metres. The light waves emitted by the electron bunches will line up in place and amplify themselves through superposition. As the electrons oscillate down the undulator they are also interacting with the magnetic components of the light they themselves are generating. The transverse component of the electrons oscillation and the magnetic field component pf the emitted light create a Lorentz force which acts upon the electrons and moves them along axial directions. This leads to the electrons creating microbunches inside their conventional bunches [1].

2.1.2. Undulators

2.2. Pump-Probe-Principle

The principle of pump-probe-spectroscopy is sending two lasers at the same object with a certain time delay so that the first laser, an optical laser, can excite the system in some way and the second laser, an X-ray pulse can measure the response. In a FEL such as the European XFEL this can be done with ultrafast X-ray pulses on a timescale of a few femtoseconds (10^{-15} s). Variating the time delay allows for a series of sequential snapshots of the process observed and variating the wavelength of the excitation laser allows for an investigation of different processces [4].

2.2.1. The Pump-Pulse

When absorbing the energy of a photon, the energy of an atoms increases as an electron in promoted into an excited energy level.

2.2.2. The Probe-Pulse

2.3. The investigated Molecule

2.3.1. Electronic Structure of $[Fe(bipy)_3]^{2+}$

2.3.2. Photophysics

3. Experimental Setup

The entrire experiment is conducted in VLab, a software modeled after a hutch in the experimental hall of the European XFEL in Schenefeld. The European XFEL is 3.4 km long and runs from the DESY research center to the European XFEL campus. A 1.9 km long superconducting linear accelerator feeds electrons with energies up to 17.5 GeV into a branched tunnel system with two 212 m long undulators that generate hard X-radiation in the range of 5 keV to 25 keV and one 125 m long undulator that generates soft X-radiation in the range of 0.25 keV to 3 keV.

VLab recreates the Femtoseconf X-ray Experiment research platform.

Datenstruktur (beam imaging and xray eye): 420x420 pixel, bei biu screen dim von 10mm x10mm, bei xray eye screen 2mm x 2mm

For further elaboration see Appendix B.

3.1. Undulator and Beamline

3.2. Monochromator and Focusing Optics

monochromator focusing optics: mirrors, power slits, solid attenuators, refractive Lenses, X-ray eye

3.3. Spectrum Analyser

diffraction gratings

3.4. Time-arrival Detector

3.5. Sample Environment

3.6. Laser System

3.7. Von Hamos Spectrometer

Quelle Versuchsmappe:[1]

4. Experimental Procedure

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

Before doing anything on a beam line it's important to characterize it.
As the absorption edge for the Fe K$\beta_{1,3}$ line of 7065 eV, the energy of the xray beam is set to 7400 eV, to avoid the ripples near the reaction (siehe **??**) but without too strong of a loss in intensity. To measure

the initial X-ray beam the power slits are fully opened and the following attenuator foils are used to bring the beam to an intensity just below saturatuion of the beam imaging unit 2 (BIU2):

The size and position of the x-ray is now measured using BIU2. Now the four-bounce crystal-monochromator is turned on and then off again to record the reduction of the spectral bandwidth. Then the dispersive single-shot spectrum analyser is used to record the spectrum of the pink X-ray beam with a bragg angle of 24°.

For the following experiment the monochromatic beam is used as it leads to a higher resolution than the pink x-ray, even though the intensity of the beam suffers.

The last step of diagnosing the beam line is using the refractive beryllium x-ray lenses to focus the beam at the imaging unit x-ray eye. It's important to appropiately adjust the attentuator foils as to not burn the imaging unit with a focussed beam.

With beryllium lenses 1, 2 and 5 (taken from ??) and the attentuator foils listed here ??, a beam with the FWHM of 13 um (siehe ??) is achievd. The total focus length is calculated with $\delta \approx 7 \cdot 10^{-6}$ [?] and the properties of the lenses themself taken from ??:

$$F = \frac{R}{2\delta N} \Leftrightarrow \frac{1}{F_{ges}} = \frac{2\delta N_1}{R_1} + \frac{2\delta N_2}{R_2} + \frac{2\delta N_5}{R_5}$$
$$\Leftrightarrow F_{ges} = \frac{1}{2\delta(\frac{N_1}{R_1} + \frac{N_2}{R_2} + \frac{N_5}{R_5})} \approx 4.76\,\mathrm{m}$$

—Notes—

X-ray energy $E_0 = 7400eV$ (has to be a good bit away from reaction edge of 7065 keV to be stable)

Alignment: attenuators to not burn through out screens (here: 4: B4C 0.8, 3: C 0.4, 2: Si 0.05, C 0.1) size and position of the beam plots (FWHM) (jitter neu!) pink vs mono beam 2D/3D plots

optical resolution: 25 um/px

Characterization: spectrum of the pink vs mono beam plots myb noch ex 7+8, aber idk

Focusing: plots of the focused beam size and position (FWHM)

4.1.2. Preparatory Estimates for the Experiment

The transition metal complex $[Fe(bipy)_3]^{2+}$ to be studied is obtained through dissolving the salt $[Fe(bipy)_3]Cl_2$ in deionized water to the desired concentration. As the actual sample is a powder consisting of small microcrystals that contain incorporated water which remained in the crystal lattice after the crystallization process.

A minimum volume of 50 ml is required to operate the liquid jet in the experiment. To ensure that the sample is completely refreshed between probe pulses, the jet needs to have a certain flow speed with an assumed repition rate of the X-ray pulse train of 1.1 MHz and an assumed top hat beam profile of the X-ray laser with the dimension 10 µm × 10 µm. The minimum required flowspeed is calculated as follows:

$$\Delta x_{\text{travel}} \geq w \quad \Rightarrow \quad v_{\text{min}} \cdot \Delta\tau \geq w \quad \Rightarrow \quad v_{\text{min}} \geq \frac{w}{\Delta\tau} \Rightarrow v_{\text{min}} = 11\,\mathrm{m\,s^{-1}}$$

With $w = 10\,\mu\mathrm{m}$ being the spatial width and $\Delta\tau = \frac{1}{f} = \frac{1}{1.1\times10^6\,\mathrm{Hz}} \approx 9.09\times10^{-7}\,\mathrm{s}$ the characteristic time.

There are three laser wavelengths available for excitation: 355 nm, 400 nm and 515 nm. To evaluate which fits the experiment best, the concentration c of the solution for the three wavelengths for a desired optical density of $OD = 1$ and a liquid jet thickness of $d = 25\,\mu$m. The optical density is defined as $OD = \epsilon \cdot c \cdot d$ with $\epsilon(\lambda)$ the wavelength dependent extinction coefficient of $[Fe(bipy)_3]^{2+}$ in water. The extinction coefficients can be estimated to be $\epsilon(355\,\mathrm{nm}) = 12\,000\,\mathrm{M^{-1}\,cm^{-1}}, \epsilon(400\,\mathrm{nm}) = 4000\,\mathrm{M^{-1}\,cm^{-1}}, \epsilon(515\,\mathrm{nm}) = 8000\,\mathrm{M^{-1}\,cm^{-1}}$ as taken from Figure 1.

The following concentrations were calculated:

$$c_{355} = 0.033\,\mathrm{mol/L}, c_{400} = 0.100\,\mathrm{mol/L}, c_{515} = 0.050\,\mathrm{mol/L}$$

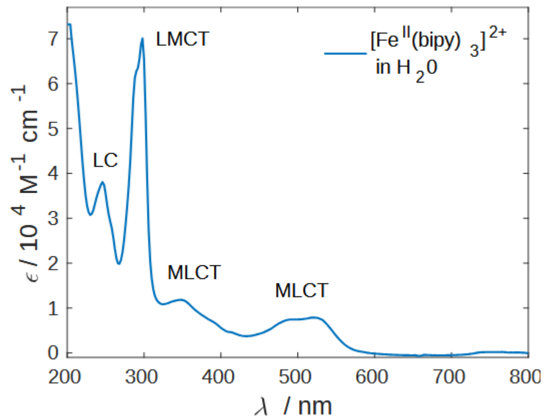

Figure 1: UV/VIS absorption spectrum of an aqueous $[Fe(bipy)_3]Cl_2$ solution. The different CT transitions are indicated. The measured molar absorptivity ϵ values lie in the expected range of $10^{-4}\ \mathrm{l\,mol^{-1}\,cm^{-1}}$ to $10^{-5}\ \mathrm{l\,mol^{-1}\,cm^{-1}}$ for dipole-allowed transitions[4]

There are different pulse energies available for each excitation wavelength: $5\,\mu$J(355 nm), $15\,\mu$J(400 nm) and $50\,\mu$J(515 nm). With that in mind the wavelangth of 400 nm was chosen, as it has a higher pulse energy, even though it's not ideal when looking at the low absorption rate. The extinction coefficien of the 515 nm is slightly better but with the low pulse energy it will be hard to see anything.

As we want to ensure that the X-ray beam always hits excited molecules we need to determine the diameter of the optical laser beam as well as the diameter of the X-ray beam correctly in relation to each other. If the optical laser and the X-ray laser had the same diameter, the x-ray would also investigate molecules that were excited only weakly or even not at all, as the optical laser has an approximately Gaussian beam shape. Ideally, the X-ray is positioned exactly in the centre of the ca. 10 times bigger optical laser beam. Then it can investigate a maximal volume of excited molecules. This also compensates for the fact that the optical laser jitters in the focus layer of the X-ray due to not being focussed there itself.
These estimates have consequences for our estimate of the needed flow speed of the liquid jet which needs to be adjusted to the new diameter of the optical laser to be 10 times faster.

Ideally, the number of absorbed optical laser photons should be less than or approximately equal to the number of sample molecules in the laser-excited volume. This reduces the likelyhood

Fun Facts: molar mass: $M([Fe(bipy)_3]^{2+}) \approx M([Fe(bipy)_3]) = 524,409\frac{g}{mol}$ concentration of the sample: $c = 0,380\frac{mol}{L}$ total attenuation of X-rays: attenuation comparison plot flow speed of the jet: $v_{jet} = 11\frac{m}{s}$

Laser: excitation wavelength: $\lambda_{pump} = 400nm$ (dazu alle fun facts mit MLCT Zuständen und so, und nicht overshooten lol) focussing, diameter of the focus Overfilling: beam sizes of the optical laser and the X-ray pulse relative to each other (plot beam diameter)

Spectromter: Van Hamos Spectrometer vs Johann Spectrometer (plot of the two geometries) crystal selection: $Si(531)$ vs $Ge(531)$ (plot of the two crystal reflections)

4.2. Performing the Experiment with Data Acquisition

5. Analysis and Interpretation

6. Discussion of the physical meaning

7. Conclusion and Outlook

References

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[2] A. Owens, "Synchrotron light sources and radiation detector metrology," *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, vol. 695, pp. 1–12, 2012.

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Appendix C Code